

The Perils of Bilateral Sovereign Debt

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Official Sovereign Debt

- A large share of sovereign borrowing takes the form of **official** debt
... Multilaterals, development banks, other governments
- Emergence of new bilateral creditors **outside** the Paris Club ▶ IDS data
... with claims to **seniority** and sometimes **confidential** terms

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- How does the presence of a large senior lender affect sovereign debt markets?
- What are its welfare implications for borrowing governments?

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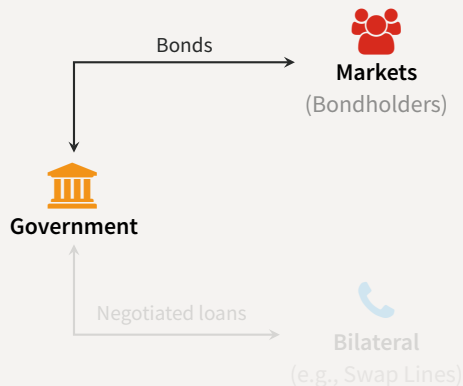
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Evaluating Senior Official Creditors

Quantitative sovereign debt model with

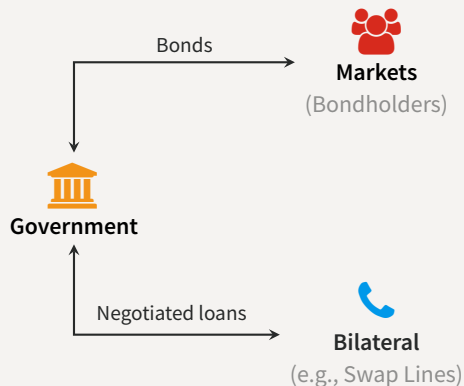
- Competitive creditors in private **markets**
- Large **bilateral** lender
 1. Superior enforcement
[de-facto seniority]
 2. Bargained terms
[price and quantity]
 3. Short-maturity loans
- Prime example: Central Bank **swap** lines
(Horn et al., 2021)
- Focus on the **interaction** between both funding sources



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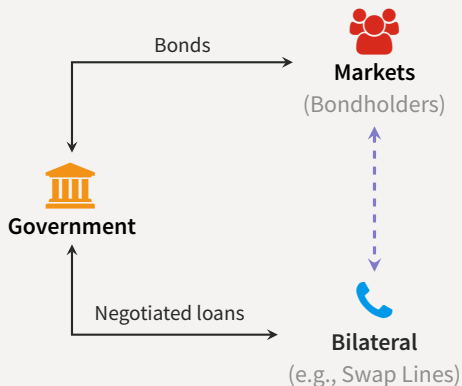
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Relational Overborrowing

Main findings

- Bilateral loans have significant effects on equilibrium outcomes
 - ... provide funding when other sources dry up (e.g. because of default risk) ▲
 - ... can also incentivize more **risk-taking** ▼
- If the rate on bilateral loans is decreasing in *market* debt [cross-elasticity]
 - ... government issues market debt more quickly, delevers more slowly
 - ... spends longer in the risky region
 - ... defaults more frequently
 - ... **welfare losses for the government**
- Cross-elasticity can emerge endogenously from **bargaining** ☞
 - ... at plausible values for bargaining weights
 - ... increased frequency of defaults dominates extra liquidity
 - ... **relational overborrowing**

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- Sovereign debt/default with interactions from ‘official’ debt
 - ... senior debt (Hatchondo, Martinez & Önder 2017), senior debt with conditionality (Boz 2011, Fink & Scholl 2016), bailout agencies (Corsetti, Guimarães & Roubini 2006, Kirsch & Rühmkorf 2017, Roch & Uhlig 2018), official debt (Dellas & Niepelt 2016, Arellano & Barreto 2024, Liu, Liu & Yue 2025)
- Data on new official creditors
 - ... Horn, Reinhart & Trebesch 2021a, 2021b, Gelpern et al. 2021, Horn, Parks, Reinhart & Trebesch 2023
- Central Bank swap lines
 - ... among advanced economies (Bahaj & Reis 2021, Cesa-Bianchi, Eguren-Martin & Ferrero 2022), data for emerging-market borrowers (Perks, Rao, Shin & Tokuoka 2021)

Roadmap

- Theory
 - Bargaining and the Cross-Elasticity
 - The Cross-Elasticity and Debt Dilution
- Quantitative Model
 - Endogenous Bargaining
 - Programming the Large Lender
- Concluding remarks

Theory

Theory

Bargaining and the Cross-Elasticity

Bargaining Induces a Cross-Elasticity

- Two periods $t \in \{0, 1\}$, endowments $y_0 < y_1$
- Government discount factor β , lender's discount factor $\beta_L > \beta$
- **Bargaining** transfer x at $t = 0$ for repayment $x(1 + r)$ at $t = 1$, weight θ for lender

$$\max_{x,r} \mathcal{L}(x,r)^\theta \times \mathcal{B}(x,r;y_0,y_1)^{1-\theta}$$

where

$$\begin{aligned}\mathcal{L}(x,r) &= \underbrace{-x + \beta_L x(1+r)}_{\text{agreement}} \\ \mathcal{B}(x,r;y_0,y_1) &= \underbrace{u(y_0 + x) + \beta u(y_1 - x(1+r))}_{\text{agreement}} - \underbrace{(u(y_0) + \beta u(y_1))}_{\text{threat point}}\end{aligned}$$

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Bargaining Induces a Cross-Elasticity

- First-order conditions for bargaining problem

$$u'(y_0 + x) = \frac{\theta}{1-\theta} \frac{\mathcal{B}(x, r; y_0, y_1)}{\mathcal{L}(x, r)}$$

Surplus sharing

$$u'(y_0 + x) = \frac{\beta}{\beta_L} u'(y_1 - x(1+r))$$

Euler

- Can prove that

$$\frac{\partial x}{\partial y_0} \leq 0 \quad \text{and} \quad \frac{\partial r}{\partial y_0} \leq 0$$

- Later** y_0, y_1 are given by market debt dynamics, so
 - Recent issuances: increases y_0 and decreases y_1 , makes you **stronger**
 - Legacy debt: lowers y_0 and y_1 , typically makes you **weaker**

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Theory

The Cross-Elasticity and Debt Dilution

Debt Dilution with Undefaultable Loans

Issue long bonds b at $t = 0$, short bonds d and loans m at $t = 1$, repay at $t = 2$

$$c_0 = q(b + d)b \quad c_1 = q(b + d)d + \phi(b, d, m)m \quad c_2(z) = y(z) - m - \min\{h(z), b + d\}$$

with

$$q(b + d) = \mathbb{P}(b + d \leq h(z)) \quad \text{and} \quad m \leq \bar{m}$$

to maximize $\sum_{t=0}^2 \mathbb{E}[u(c_t)]$

Commitment: Choose d internalizing effect on initial prices

$$u'(c_0) \underbrace{q'(b + d)b}_{\text{past prices}} + u'(c_1) \left(\underbrace{q(b + d)}_{\text{revenue}} + \underbrace{q'(b + d)d}_{\text{current prices}} + \underbrace{\phi'_d(b, d, m)m}_{\text{cross-elasticity}} \right) = q(b + d) \mathbb{E}[u'(c_2^R)]$$

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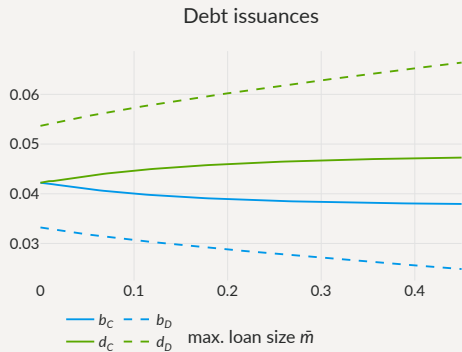
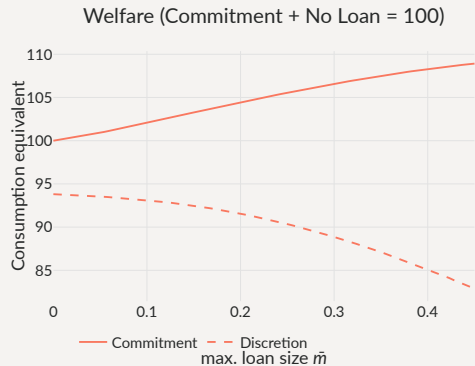
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not under discretion

Debt Dilution with Undefaultable Loans and Cross-Elasticity



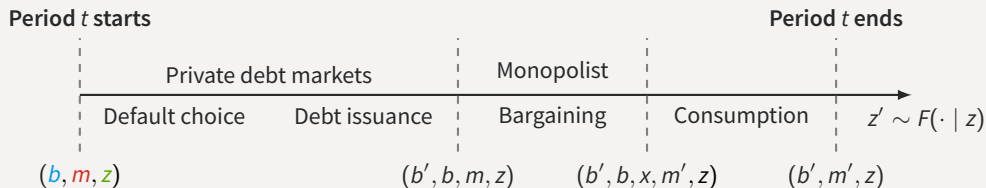
Takeaways

- Bargaining can induce a cross-elasticity from market debt to loan terms
 - ... through threat-point manipulation
- The cross-elasticity worsens the debt dilution problem
 - ... it can also create welfare losses if strong enough

Quantitative Model

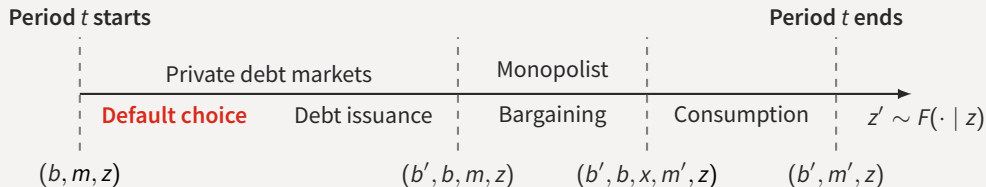
Timeline of Events

- Enter period t owing b to bondholders, m to monopolist, income $y(z)$



Timeline of Events

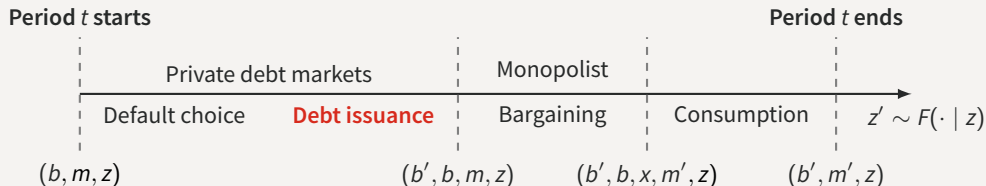
- Choose to **repay** or **default** the *market* debt subject to convex output costs



$$v(b, m, z) = \max \{ v_R(b, m, z) + \epsilon_R, v_D(m, z) + \epsilon_D \}$$

Timeline of Events

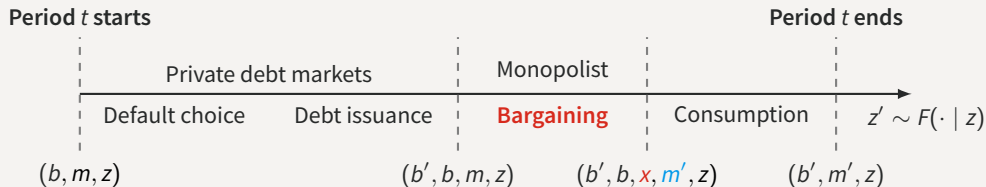
- If repaid, issue new (long-term) debt b' in markets at price q [coupon rate κ , maturity $1/\delta$]



$$q(b', b, m, z) = \beta_L \mathbb{E} [(1 - 1_{\mathcal{D}}(b', m', z')) (\kappa + (1 - \delta)q(b'', b', m', z')) | z]$$

Timeline of Events

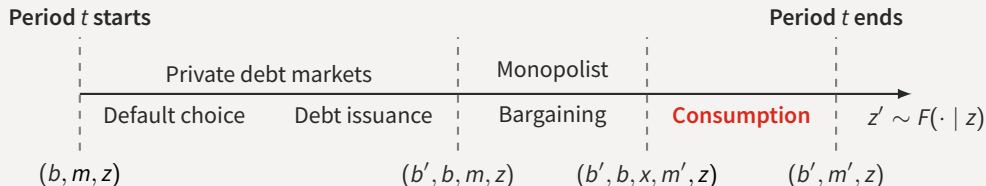
- Meet with senior lender, decide any transfers x and new/remaining balance m'



$$r(x, m', m) = \frac{m'}{x + m} - 1$$

Timeline of Events

- Consume **output** plus **revenues from debt issuance** plus **transfers** minus **debt service**



$$c_R = y(z) + q(b', b, m, z)(b' - (1 - \delta)b) + x_R(b', b, m, z) - \kappa b$$

$$c_D = y(z) - h(z) + x_D(m, z); \quad x_D(m, z) \leq \Gamma(m)$$

Calibration

- Standard strategy in sovereign debt: match Argentina 1993-2001 to model w/o swaps

	Parameter	Value
Sov's discount factor	β	0.9607
Sov's risk aversion	γ	2
Pref. scale	χ	0.025
Lender's barg. power	θ	0.5
Risk-free rate	r	0.01
Duration of debt	ρ	0.05
Income autocorr.	ρ_z	0.9484
Std. dev. y_t	σ_z	0.02
Reentry prob.	ψ	0.0385
Default cost: linear	d_0	-0.2514
Default cost: quadratic	d_1	0.292

	Data	Only market
Avg spread (bps)	815	740
Std spread (bps)	443	485
Debt-to-GDP (%)	17.4	18.1

Note moments from pre-default samples

Quantitative Model

Endogenous Bargaining

Bargaining Stage with Monopolist

- At state z , owing debt b bonds and m on the loan and having issued b'

$$\max_{x, m'} \mathcal{L}_R(b', x, m, m', z)^\theta \times \mathcal{B}_R(b', b, x, m, m', z)^{1-\theta}$$

Government surplus
Lender surplus

- Lender's surplus

$$\mathcal{L}_R(b', x, m, m', z) = \underbrace{(-x + \beta_L \mathbb{E}[h(b', m', z') | z])}_{\text{agreement}} - \underbrace{(m + \beta_L \mathbb{E}[h(b', 0, z') | z])}_{\text{threat point}}$$

- Government's surplus

$$\begin{aligned} \mathcal{B}_R(b', b, x, m, m', z) = & \underbrace{u(y(z) + B(b', b, m, z) + x) + \beta \mathbb{E}[v(b', m', z') | z]}_{\text{agreement}} \\ & - \underbrace{(u(y(z) + B(b', b, m, z) - m) + \beta \mathbb{E}[v(b', 0, z') | z])}_{\text{threat point}} \end{aligned}$$

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same sdf as markets

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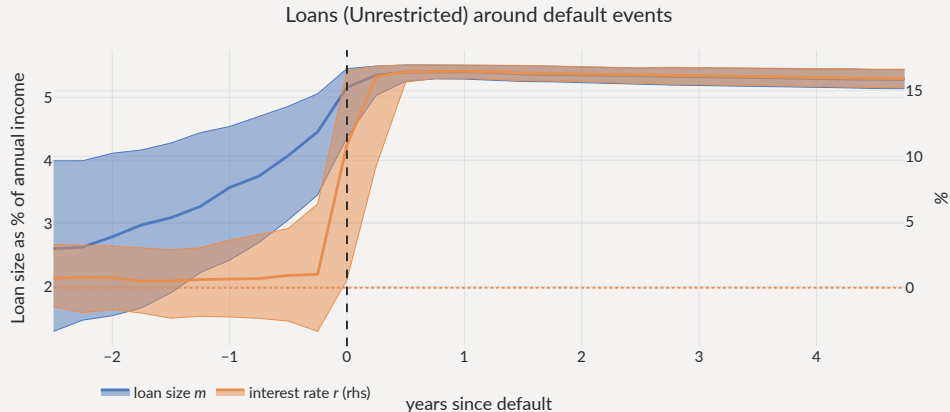
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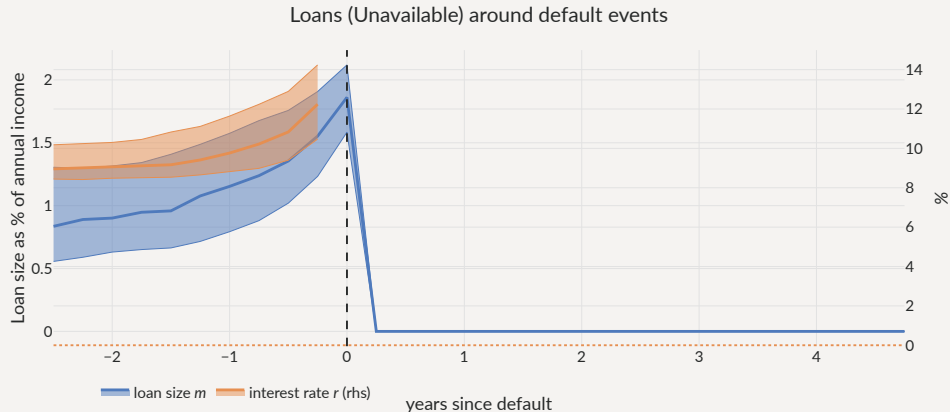
Usage of Loans

- Sharp acceleration before defaults, maxed out *in* default if possible



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Limiting Loans in Default

• **Limited:** cannot borrow in default $\Gamma(m) = m$

Unavailable: $\Gamma(m) = 0$

	Only market	Unrestricted, $\theta = 0.5$	Limited, $\theta = 0.5$	Unavailable, $\theta = 0.5$
Avg spread (bps)	630	1,398	1,209	650
Std spread (bps)	474	894	860	381
$\sigma(c)/\sigma(y)$ (%)	104	99	100	104
Debt to GDP (%)	16.4	17.6	17.6	18
Loan to GDP (%)	0	2.44	2.22	0.903
Loan spread (bps)	–	-298	-126	523
Corr. loan & spreads (%)	–	68.9	67.7	63.8
Default frequency (%)	5.03	9.55	9.04	5.28
Welfare gains (rep)	–	-0.095%	-0.038%	0.12%

Note Statistics from each ergodic distribution

Limiting Loans in Default

- Default rates:  with loans (less with Limited/Unavailable)

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Limiting Loans in Default

• Welfare:

≈ with loans (Unavailable ≧ Limited ≧ Unrestricted)

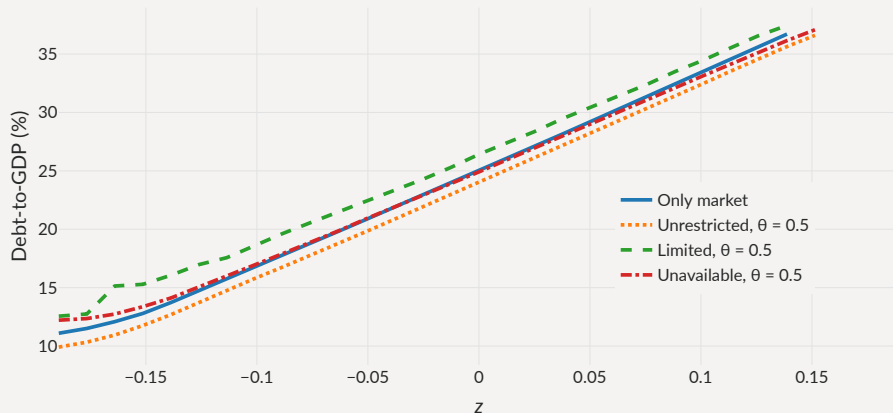
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Default Barriers with Loans

- **Unrestricted**: default barrier marginally inward, others marginally *outward*

Debt levels at which $P(b,m,z)$ crosses 50%

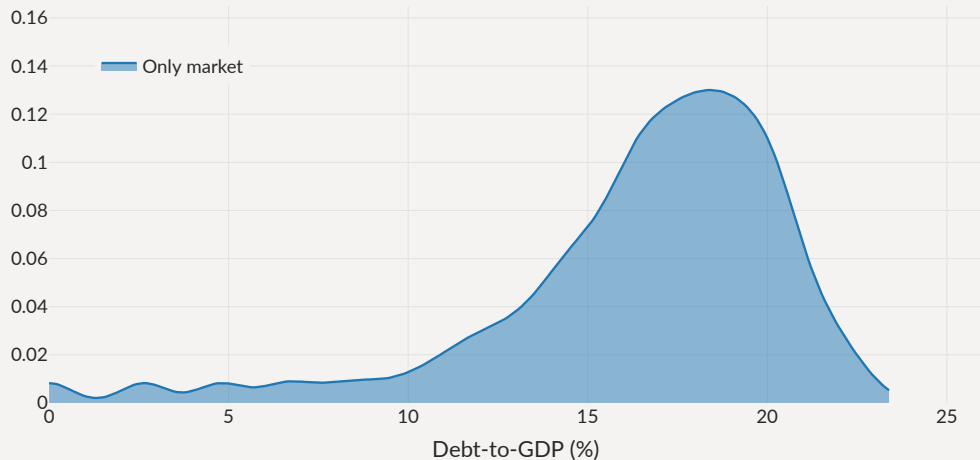


If Loans help repay the debt,

Why are there **more** defaults with loans?

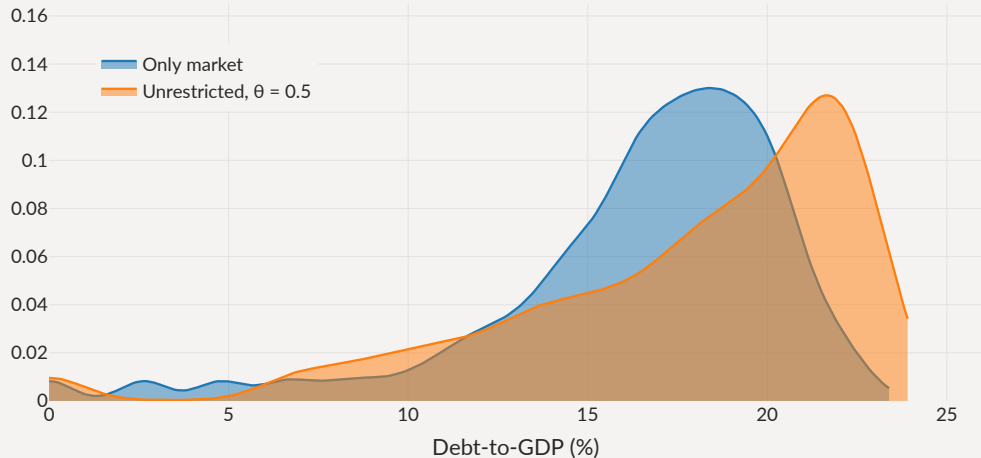
Debt Levels with Loans

Distribution of debt levels



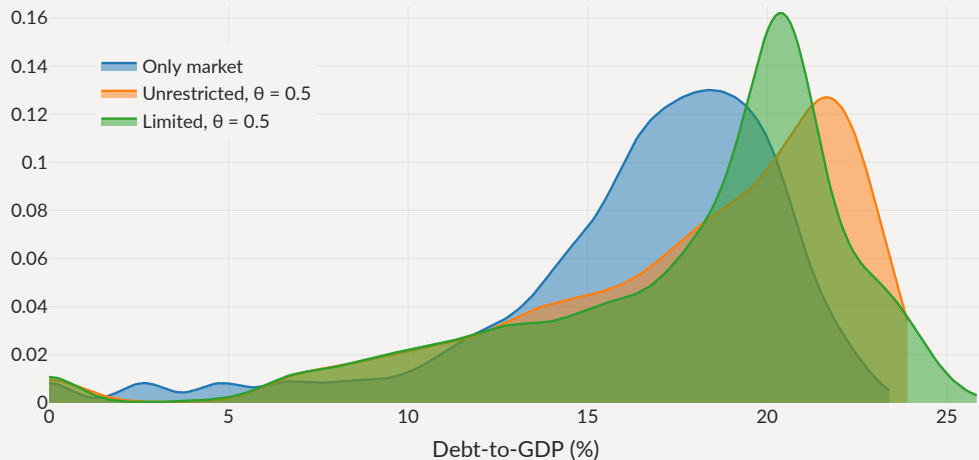
Debt Levels with Loans

Distribution of debt levels



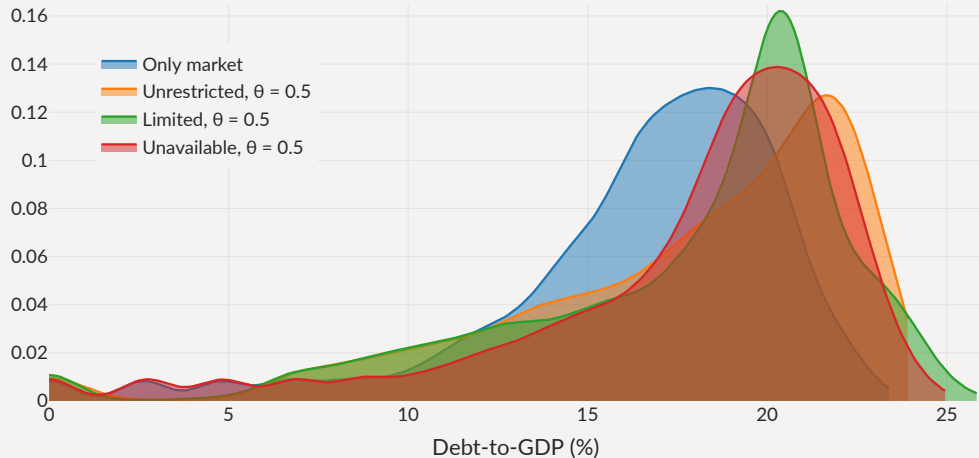
Debt Levels with Loans

Distribution of debt levels



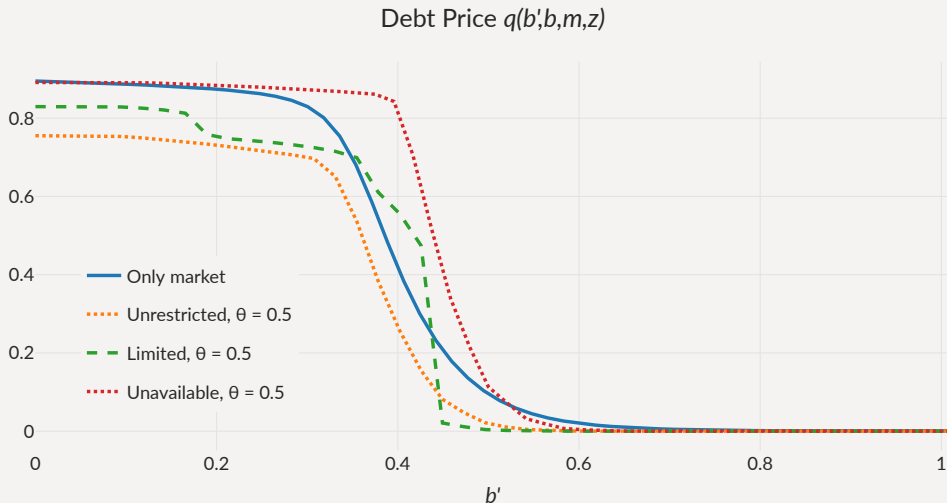
Debt Levels with Loans

Distribution of debt levels



Debt Prices with Loans

Lower prices with same ex-post default rates: **relational overborrowing** similar to debt dilution

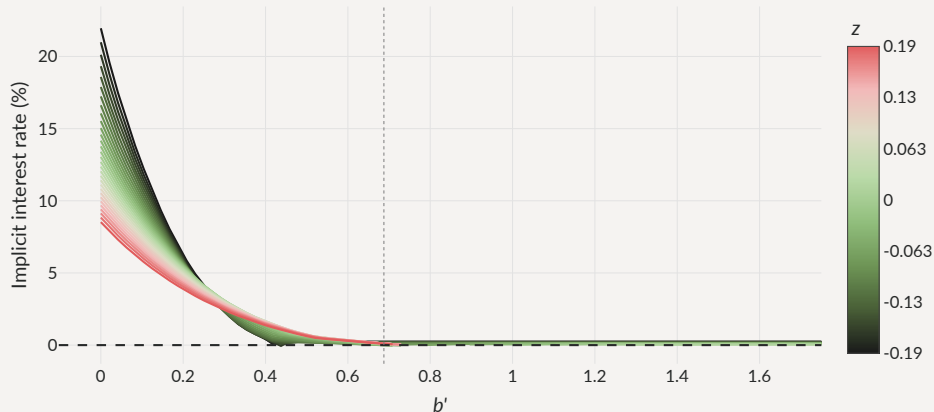


Government's surplus

$$\begin{aligned} \mathcal{B}_R(b', b, x, m, m', z) = & u(y(z) + B(b', b, m, z) + x) + \beta \mathbb{E} [v(b', m', z') \mid z] \\ & - (u(y(z) + B(b', b, m, z) - m) + \beta \mathbb{E} [v(b', 0, z') \mid z]) \end{aligned}$$

- Revenues from debt issuance $B(b', b, m, z)$ modulate the value of the threat point
 - After large revenues (high q , high b'), gov't flush with cash, **strong** in bargaining
 - After bad issuance (low q or low b'), gov't **weak** in bargaining

Threat point manipulation: increase market b' to reduce bilateral r **Cross-elasticity**
Loan interest rate (Unavailable)



Government's Euler equation for bonds

$$u'(c) \left(q + \frac{\partial q}{\partial b'} i + \frac{1}{1+r} \frac{\partial m'}{\partial b'} + \frac{\partial \frac{1}{1+r_b}}{\partial b'} m' \right) = \beta \mathbb{E} \left[v_b(b', m', z') + v_m(b', m', z') \frac{\partial m'}{\partial b'} \mid z \right]$$

- An extra bond:

... yields marginal revenue $q + \frac{\partial q}{\partial b'} i$ [standard debt-Laffer curve effect]

... makes you **stronger** in bargaining now

... affects the amount m' borrowed from monopolist

... affects terms of borrowing with monopolist

... makes you **weaker** in bargaining later

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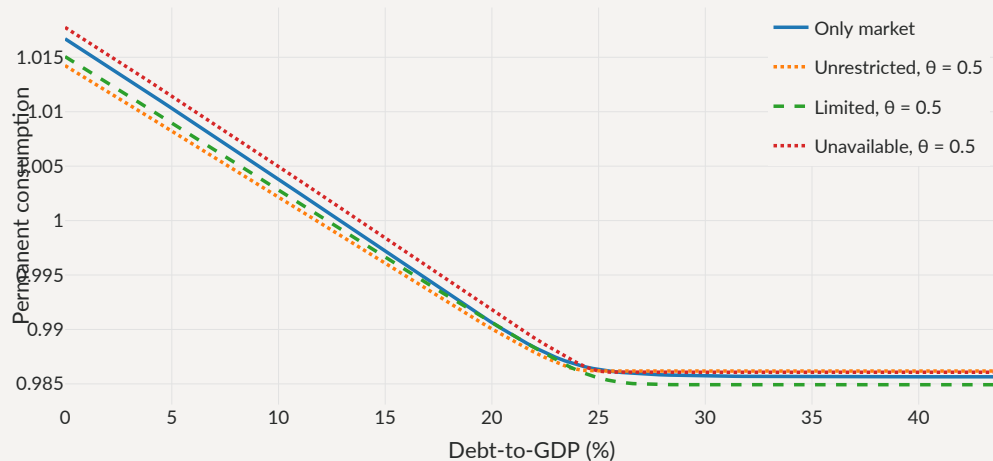
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Welfare Effects of Bilateral Loans

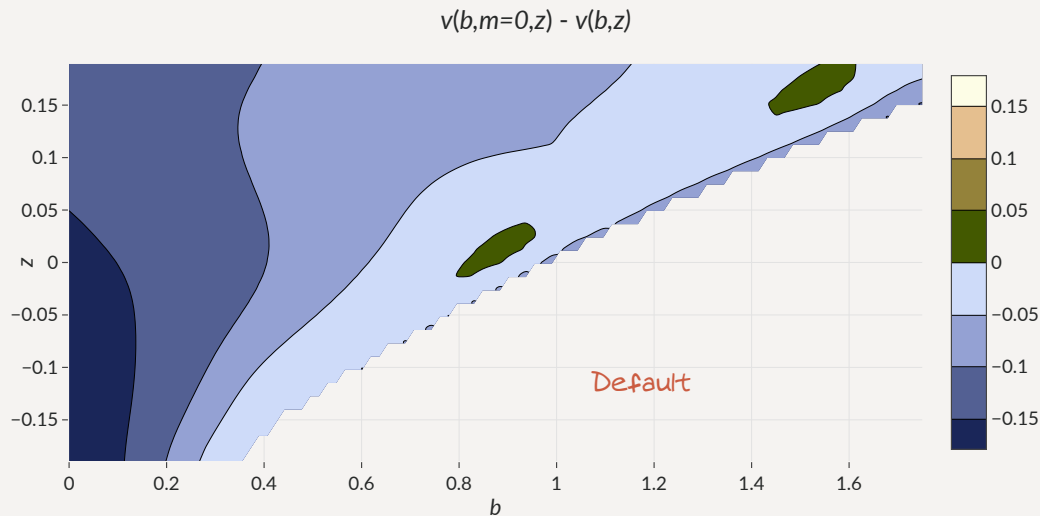
Limited $\succcurlyeq$ Unrestricted, but...

Value at $m=0$, average output



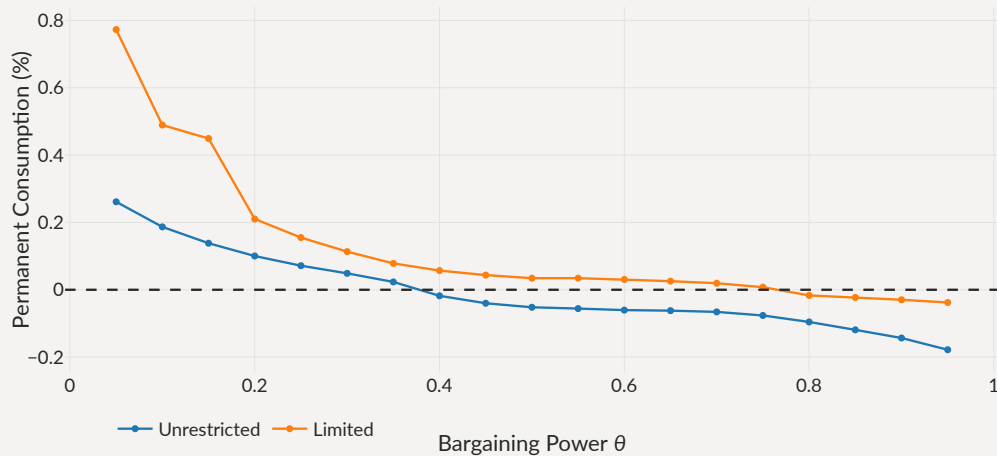
Welfare Effects of Bilateral Loans (cont'd)

Overall welfare effect small: more risk, but more frontloading in short run.



Bargaining Power and Welfare

Welfare Gains



Quantitative Model

Programming the Large Lender

Possible Rules

- Explore interest rate rules of the form

$$r(b', m') = \max\{r^*, \alpha_0 + \alpha_b b' + \alpha_m m'\}$$

- Two versions

Size-dependent

$$\alpha_0 > 0, \alpha_b = 0, \alpha_m > 0$$

Risk-inducing

$$\alpha_0 > 0, \alpha_b < 0, \alpha_m \geq 0$$

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Equilibrium with Exogenous Rules

- ‘Only market’ standard calibration to Argentina 1993-2001

	Only market	Size dependent r	Risk inducing r	Limited, $\theta = 0.5$
Avg spread (bps)	630	402	550	650
Std spread (bps)	474	247	340	381
$\sigma(c)/\sigma(y)$ (%)	104	105	104	104
Debt to GDP (%)	16.4	17.5	17	18
Loan to GDP (%)	0	0.36	0.12	0.903
Loan spread (bps)	–	704	6,535	523
Corr. loan & spreads (%)	–	51.4	1.73	63.8
Default frequency (%)	5.03	3.46	4.55	5.28
Welfare gains (rep)	–	0.25%	-0.018%	0.12%

Note Statistics from each ergodic distribution

Equilibrium with Exogenous Rules

- Default rates:  with size dependent  with risk-inducing

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Note Statistics from each ergodic distribution

Equilibrium with Exogenous Rules

- Welfare: ↑ with size dependent ↓ with risk-inducing

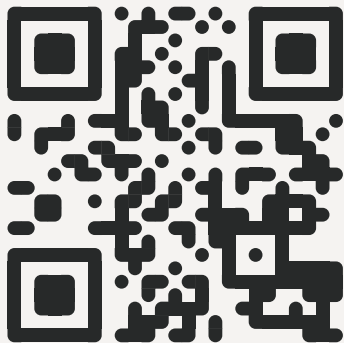
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Concluding remarks

The Perils of Bilateral Sovereign Debt

- Simple model of borrowing from **markets** and a **senior bilateral lender**
 - ... strong interaction between two markets for sovereign debt
 - ... even if bilateral loans are **not** used intensely on the equilibrium path
- **Dangerous** when bilateral interest rate responds negatively to *market* debt
 - ... cross-elasticity induces risk-taking, more defaults, welfare losses
 - ... Bargaining as an example of situation where cross-elasticity emerges
- Cross-elasticity constitutes a simple test to assess welfare gains of **new** instruments
 - ... or a boost to the gains of fiscal rules, state-contingent debt...

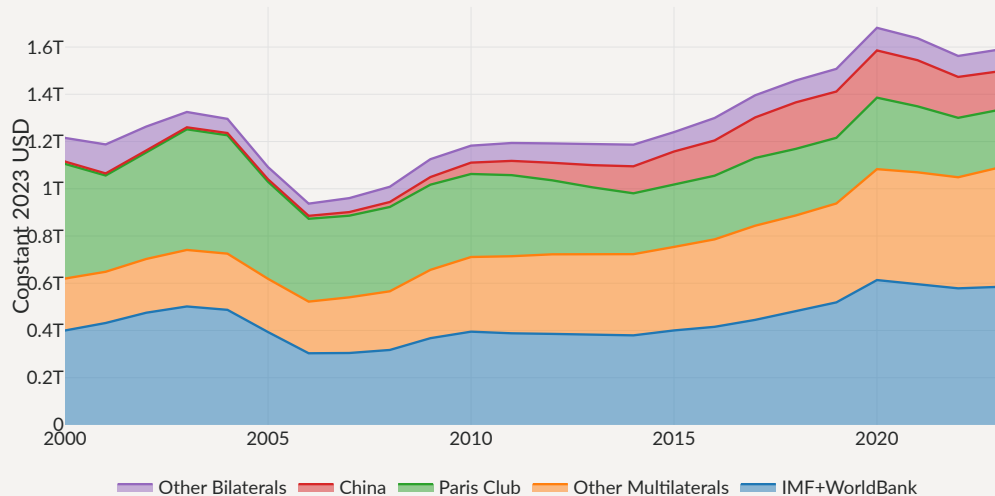


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A New Landscape for Official Sovereign Debt

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Total Official Debt



Loan drawings m' (Unavailable)

